DIGITAL TWIN CHALLENGE 2026 · TYPE 2 DIABETES

Prahari

A digital twin that warns of a glucose spike up to two hours before it happens, explains why, and simulates what would prevent it

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All patients are synthetic (Synthea EHR + simulated CGM and smartwatch). Research prototype, not a medical device.

THE PROBLEM

India's diabetes problem is a timing problem: the spikes happen after meals, the doctor sees one number every three months

101 M

adults in India with diabetes, plus 136 M with prediabetes (ICMR-INDIAB, 2023)

43%

of adults with diabetes in India are undiagnosed (IDF Diabetes Atlas, 11th ed.)

62%

of daily energy in Indian diets comes from carbohydrate (Anjana et al., 2025)

1 number

HbA1c, every few months, is what most doctors see. It says nothing about when or why

What is missing

- CGMs and smartwatches now record every 5 minutes, but the data sits in separate apps, away from the medical record
- No one turns it into a warning early enough to act: a walk, a smaller portion, a call from the doctor
- Doctors have no way to test "what if she walked after lunch?" for one specific patient

THE IDEA

Prahari gives the doctor four things today's care cannot: a warning, a reason, a personal model and a what-if

Warn

Probability of a spike above 180 and above 250 mg/dL in the next 2 hours, every 15 minutes, with a 2-hour forecast and an 80% band

Explain

The top three reasons behind each alert in plain words (SHAP): the meal just logged, the twin's predicted peak, a short night

Personalise

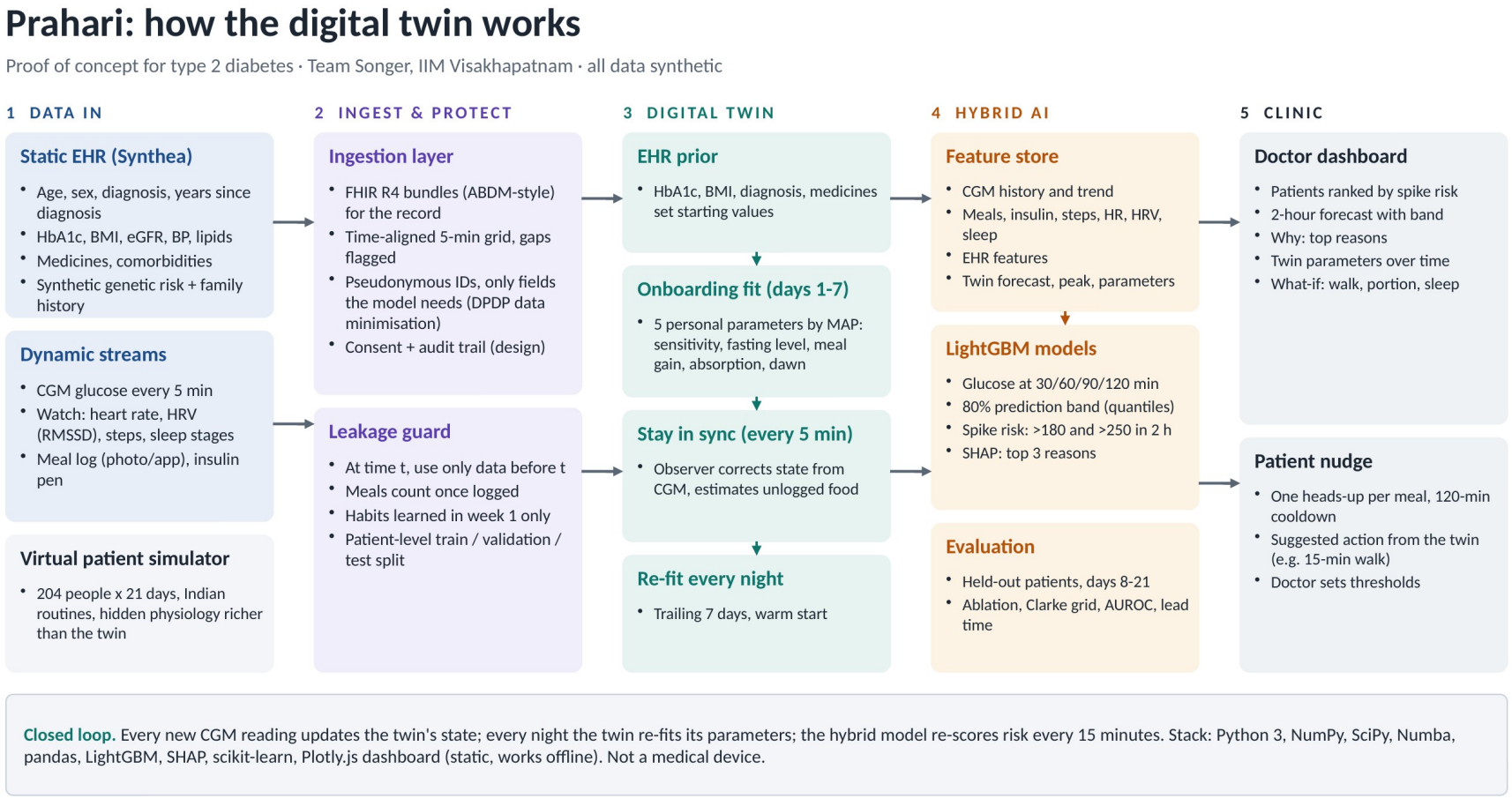
A twin per patient: insulin sensitivity, fasting level, meal response, absorption speed, dawn effect, re-fitted every night

Act

What-if in the browser: walk 15 minutes, smaller portion, rice or chapati, a short night. At most one nudge per 2 hours for the patient

ARCHITECTURE

Five layers: data in, ingest and protect, the digital twin, hybrid AI, and the clinic



DATA

204 synthetic people lived 21 days of Indian routines, with a hidden physiology the twin never sees

Static record (Synthea)

- 94 with type 2 diabetes, 110 with prediabetes; 107 women
- Age, years since diagnosis, HbA1c, BMI, eGFR, BP, lipids
- Metformin, premixed insulin (18 users), comorbidities
- Synthetic family history and polygenic risk, labelled as synthetic
- HbA1c values that contradict the diagnosis are repaired and flagged

Daily life (simulated)

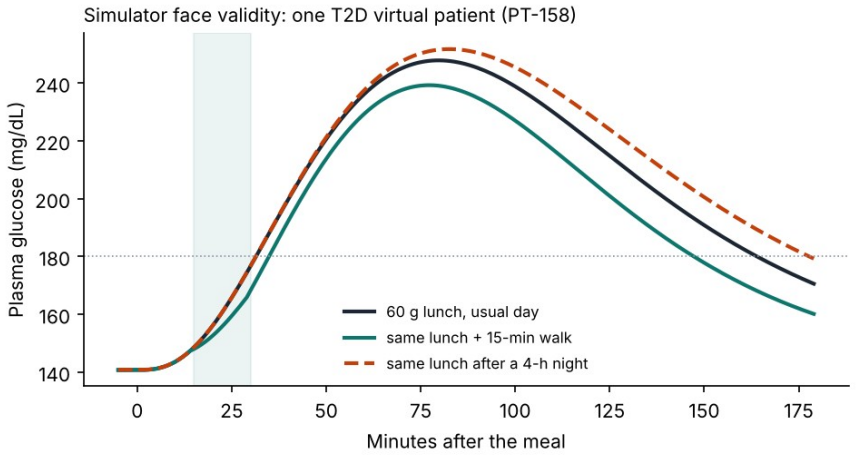
- North or south Indian meals: rice, chapati, idli, dosa, poha, biryani, chai with sugar
- 16,354 meals; 76% logged, with carb-counting error
- Walks, commutes, short nights, stressful work days
- Insulin before breakfast and dinner

What the devices record

- CGM every 5 minutes, sensor error 9.5% MARD, gaps, sensor change on day 11
- Watch: heart rate, HRV (RMSSD), steps, sleep stages
- 1.23 million 5-minute time points
- A richer hidden physiology produces the truth; the twin is simpler, so it must cope with mismatch

The virtual patients respond the way published studies say real patients do

Check	Simulator	Published
15-min walk after a meal	-16% post-meal glucose	-12% overall, -22% after dinner (Reynolds 2016)
One 4-hour night	sensitivity -25%, +10% breakfast rise	about -25% glucose infusion rate (Donga 2010)
Dawn rise in T2D	median 12 mg/dL	median 16 mg/dL (Monnier 2013)
Mean glucose vs HbA1c	r = 0.96	r = 0.92, eAG formula (Nathan 2008)
Time in range vs HbA1c	-18 points per 1%	about -17 points per 1% (Beck 2019)
CGM error (MARD)	9.5%	8.2-9.1%, Dexcom G7 (Garg 2022)



Same lunch, three days: usual, with a 15-minute walk, and after a 4-hour night.

The twin is a living model: built from the record, fitted in week one, synced every 5 minutes, re-fitted every night

1 Prior

HbA1c, BMI, diagnosis and medicines set starting values

2 Onboard

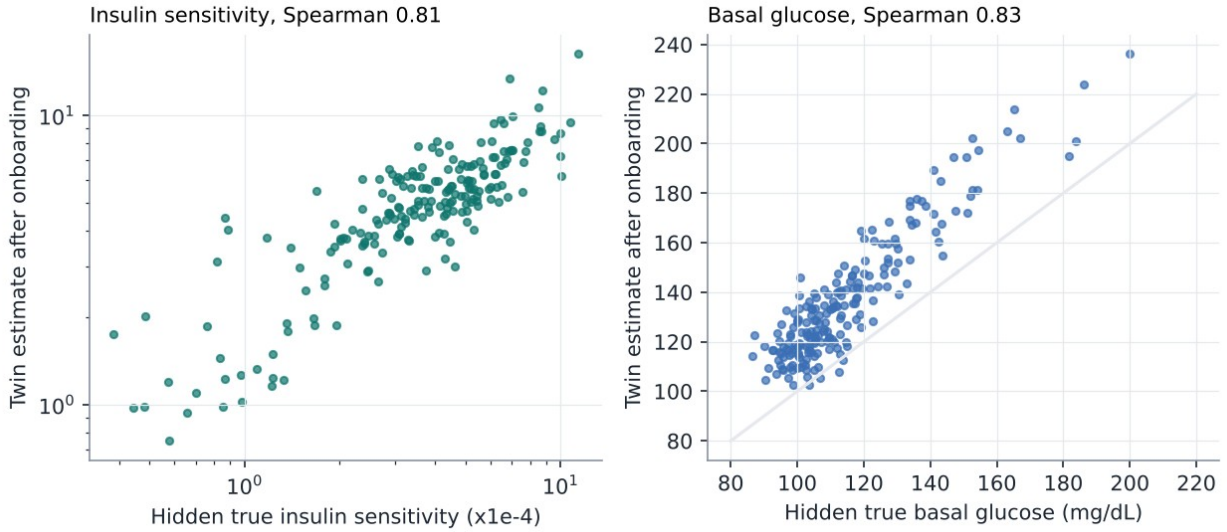
Days 1-7: five personal parameters fitted by MAP on its own forecasts

3 Sync

Each CGM reading corrects the state; estimates unlogged food

4 Re-fit

Every midnight on the last 7 days; day d only uses data before d



It recovers the hidden physiology. The record alone ranks insulin sensitivity poorly (Spearman 0.37 against the hidden truth). After week one the twin reaches 0.81, and 0.82 by day 21. Re-fitting nightly beats freezing it after week one: 60-min error 28.8 vs 30.5 mg/dL.

HYBRID AI

A hybrid model: the twin's physics plus machine learning on every data source, with no look-ahead

CGM	25 features	level, trends over 5-120 min, variability, gaps
Lifestyle	27 features	meals logged, habitual meal times from week 1, insulin, steps, heart rate, HRV vs own baseline, sleep
EHR	18 features	age, HbA1c, BMI, eGFR, medicines, comorbidities, genetic risk
Twin	17 features	its 30-120 min forecast, predicted peak and timing, personal parameters, unexplained glucose

Models (LightGBM)

- Glucose at 30, 60, 90 and 120 minutes, plus 10th and 90th percentiles for an 80% band
- Spike risk: onset above 180 (and above 250) mg/dL for at least 15 minutes within 2 hours
- Alert threshold frozen on validation patients at 70% precision; one notification per 120 minutes
- SHAP reasons for every alert

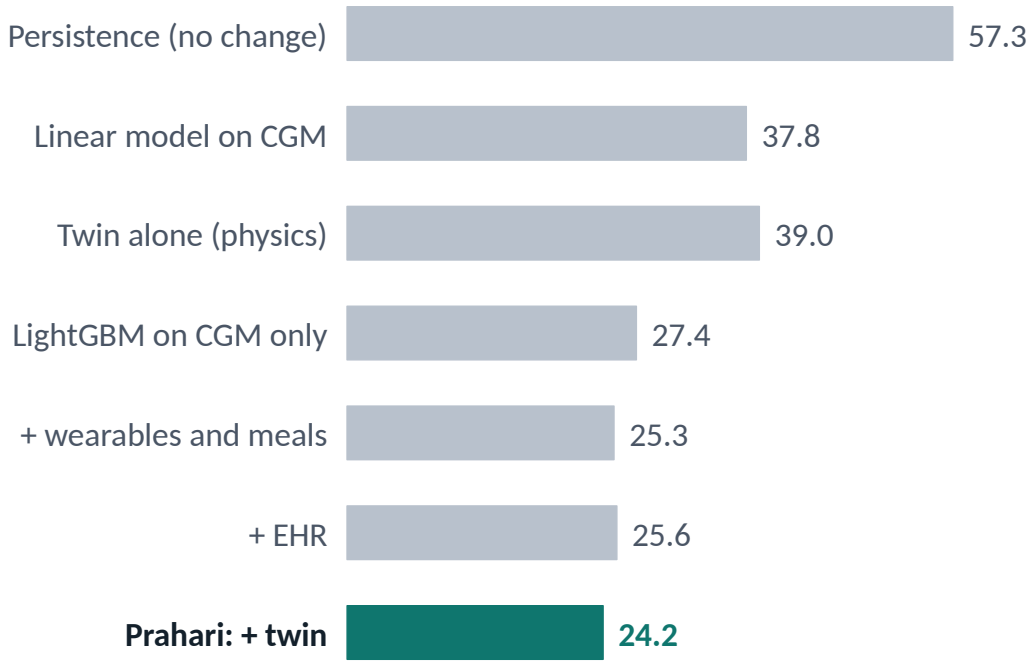
Leakage controls

- Train, validation and test are different patients (122 / 41 / 41)
- Only data before the moment of prediction; meals count once logged
- Results on days 8-21, after the twin's onboarding week

RESULTS: FORECASTING

Prahari forecasts glucose 2 hours ahead with an error of 24.2 mg/dL, less than half of 'no change'

RMSE at 120 minutes, mg/dL (41 held-out patients, days 8-21)



The twin's features cut error by 1.7 mg/dL at 60 min (95% CI 1.2-2.2) over the same model without them.

17.2 · 21.1 · 24.2

RMSE at 30 · 60 · 120 minutes (mg/dL)

41 → 25

60-min error in the 45 min after a meal: CGM-only vs Prahari

99.6%

of 60-min forecasts in Clarke zones A+B (clinically acceptable)

RESULTS: SPIKE WARNINGS

It flags 3 in 4 spikes about 85 minutes before they start; severe spikes remain the hard part

76%

of spikes above 180 flagged 15-120 min ahead

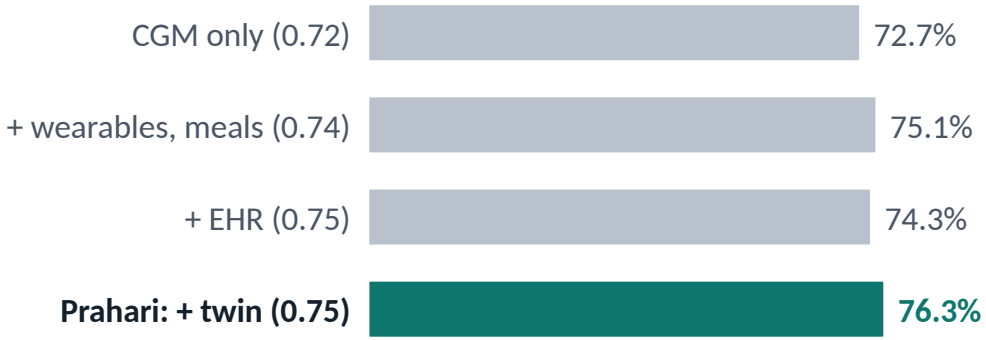
85 min

median warning before the spike starts

60%

of notifications followed by a spike (2.9 per patient-day)

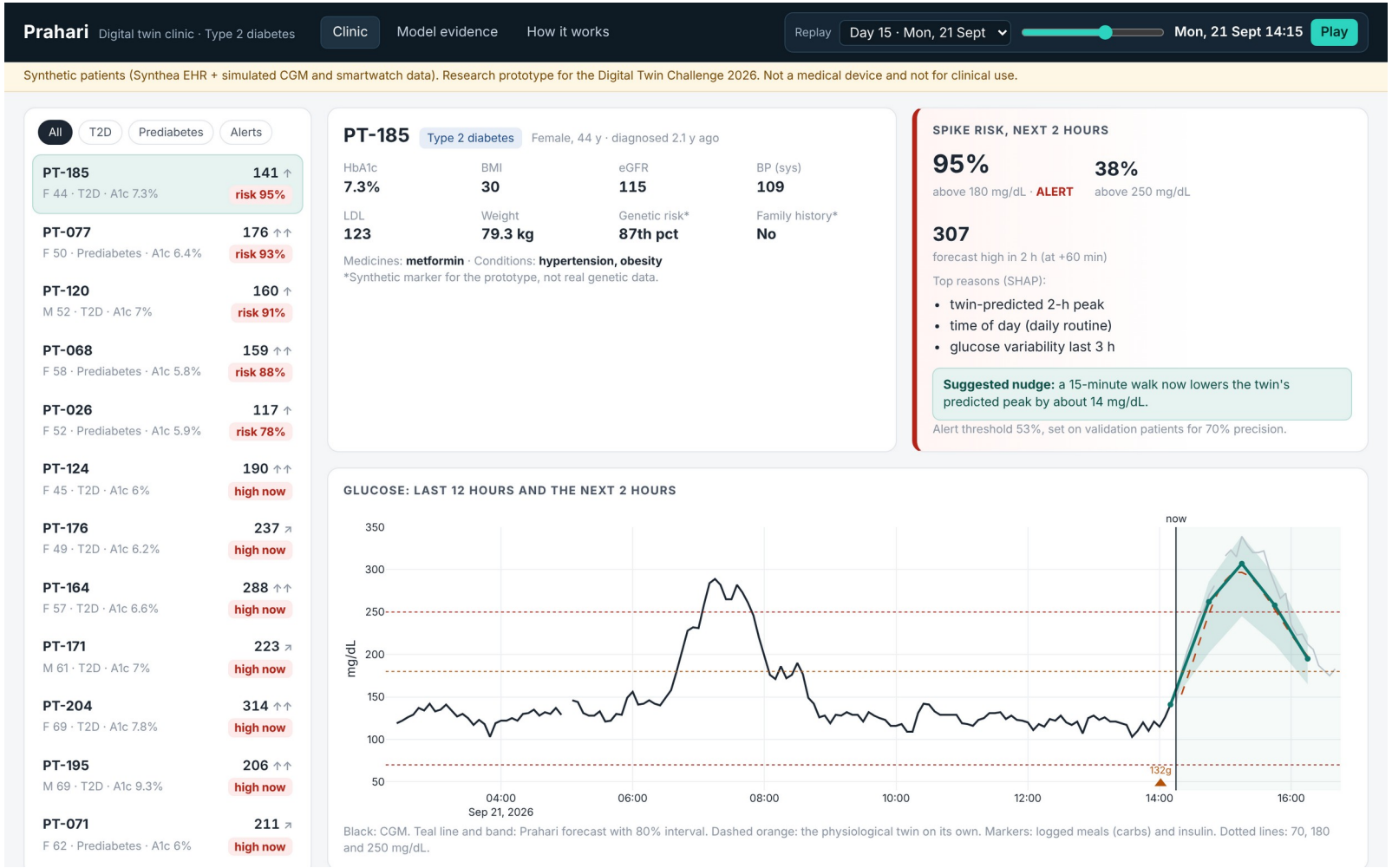
Spikes above 180 flagged early, every model at the same alert budget (AUPRC in brackets)



- The twin's physics alone flags 78%, but only 35 min ahead; learning routines stretches it to 85
- Wearables and meal logs add most to alerts; the twin adds little on top
- Severe spikes above 250: 37% flagged (CGM only 30%) at 0.5 notifications a day

CLINIC DASHBOARD

The doctor's dashboard: who needs attention now, what will happen, and why



- Patients ranked by spike risk in the next 2 hours
- Alert card: risk above 180 and 250, forecast high, top three reasons, suggested nudge
- 12 hours of CGM, meals and insulin, then the 2-hour forecast with its 80% band and the twin's own line
- Replay any moment of the test fortnight; works offline, one HTML file

USE CASE: PT-185

The twin answers what-if questions for this patient: a 15-minute walk takes 14 mg/dL off her peak

SPIKE RISK, NEXT 2 HOURS

95%
above 180 mg/dL · **ALERT**

38%
above 250 mg/dL

307
forecast high in 2 h (at +60 min)

Top reasons (SHAP):

- twin-predicted 2-h peak
- time of day (daily routine)
- glucose variability last 3 h

Suggested nudge: a 15-minute walk now lowers the twin's predicted peak by about 14 mg/dL.

Alert threshold 53%, set on validation patients for 70% precision.

44-year-old woman with T2D, HbA1c 7.3%, on metformin. Mon, 21 Sept 14:15, 14 minutes after a 132 g lunch; glucose 141. Prahari: 95% risk above 180, forecast high 307. A 15-minute walk: peak 297 → 283, time above 180 120 → 105 min.

WHAT-IF: ASK THE TWIN

Meal: Nothing new (only what is already logged)

Starts in: 0 min

Portion: 100%

Walk after the meal: 15 min

Last night's sleep: 4.9 h

The walk (100 steps/min) starts 15 min after a new meal, or at the start time if nothing new is eaten.

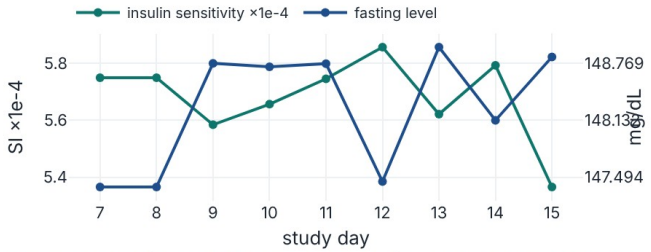
Runs this patient's calibrated twin in your browser.



Peak, usual	Peak, scenario	Minutes above 180 (4 h)
297	283	120 → 105

THE PATIENT'S TWIN

Parameter (day 15)	Twin now	EHR prior	Panel rank
Insulin sensitivity	**5.37×10^{-4}**	2.89	64th
Fasting glucose level	**149 mg/dL**	130	71st
Meal response (carb gain)	**0.65×**	1.00	21st
Absorption speed	**2.8 %/min**	3.0	71st
Dawn effect	**0.33 mg/dL/min**	0.35	71st



Parameters start from an EHR-based prior, are fitted in the onboarding week (days 1-7) and re-fitted every night on the trailing 7 days.

DEPLOYMENT IN INDIA

Built for India's health stack: FHIR records, data minisation, and a clear path through regulation

Interoperability

- Records exchanged as FHIR R4, the standard of the ABDM FHIR implementation guide (v6.5.0)
- Over 93 crore ABHA accounts already exist (PIB, July 2026) to link records

Privacy

- Only the fields the model needs are kept, no names or identifiers
- Designed for consent and purpose limits under the DPDP Rules 2025, whose main duties apply from 13 May 2027

Regulation and pilot

- As decision-support software it would be a medical device under CDSCO's risk classes A-D
- Next: validate on real CGM data, then a 12-week pilot with one diabetology clinic measuring time in range

What a pilot would measure: time in range and time above 250 mg/dL before and after 12 weeks, share of alerts the doctor found useful, false alerts per patient-day, and minutes of doctor time per patient review.

What these results do not show yet, and what I would do next

Limits

- All patients are synthetic and more regular than real people, so real-world error will be higher. On the real OhioT1DM data, published deep models reach 55-66 mg/dL at 2 hours (Tena 2021)
- At HbA1c 7% my virtual patients have more time in range than real cohorts
- No hypoglycaemia in the simulated cohort, so lows are not predicted yet
- The 80% band covers 78% of outcomes at 60 min; it needs calibration
- Many alerts come before habitual meals: the model learns routines

Next

- Validate on open real-world T2D CGM data, then a consented Indian cohort
- Add hypoglycaemia risk for insulin users and conformal prediction bands
- Estimate carbs from meal photos, so logging takes one tap
- 12-week clinic pilot: doctor dashboard plus patient nudges, measured by time in range

The bar to clear on real data: on the real ShanghaiT2DM CGM dataset, a published Bayesian model reaches 13.5 and 21.4 mg/dL at 30 and 60 minutes (Sun & Kosmas, 2024).

Prahari

A twin for every patient. An early warning for most spikes. A reason for every warning. An action the twin has simulated.

- Code, data, models and results: public GitHub repository Songer_IIMVisakhapatnam (MIT licence)
- Dashboard: docs/index.html, also on GitHub Pages
- Reproduce everything: python scripts/run_all.py

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